Assignment - 5

① (i) $A(F) = \frac{1}{2}(\bar{B} \times F)$

$\bar{B} = B\hat{k}$; $\bar{F} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore \bar{B} \times F = B[+x\hat{j} - y\hat{i}]$

$\therefore A = \frac{B}{2}[-y\hat{i} + x\hat{j}]$

$\therefore A_x = -\frac{B}{2}y$; $A_y = \frac{B}{2}x$; $A_z = 0$

$\therefore \bar{\pi} = \bar{p} - \frac{e}{c}\bar{A} = \bar{p} - e\bar{A}$ (kinetic momentum operator)

$\therefore \pi_x = p_x - eA_x$; $\pi_y = p_y - eA_y$; — (ii)

— (i)

[No z component as the particle is confined to the xy plane].

$\therefore H = \frac{1}{2m}\sum_i \pi_i^2 = \frac{1}{2m}[(p_x - eA_x)^2 + (p_y - eA_y)^2]$

$$\boxed{H = \frac{1}{2m}\left[\left(-i\hbar\frac{\partial}{\partial x} + \frac{eB}{2}y\right)^2 + \left(-i\hbar\frac{\partial}{\partial y} - \frac{eB}{2}x\right)^2\right]}$$

— (iii) (Ans)

$\pi_x = -i\hbar\frac{\partial}{\partial x} + \frac{eBy}{2}$; $\pi_y = -i\hbar\frac{\partial}{\partial y} - \frac{eB}{2}x$

(ii) $H = \frac{1}{2m}[\pi_x^2 + \pi_y^2]$

In SHO: $H = \frac{1}{2m}[p^2 + (m\omega x)^2]$

$a_\pm = \frac{1}{\sqrt{2\hbar m\omega}}[m\omega x \mp ip]$

Hence, in this problem;

$a = \frac{A}{\cancel{\sqrt{2\hbar m\omega}}}[\pi_x + i\pi_y]$; $a^+ = A^+[\pi_x - i\pi_y]$

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We can calculate A by the commutation relat^n

$[a, a^\dagger] = +1$

$[\pi_x, \pi_y] = [p_x + \frac{eB}{2} y, \; p_y - \frac{eB}{2} x]$ [from (i) & (ii)]

$= \underbrace{[p_x, p_y]}_{\to 0} - \frac{eB}{2}[p_x, x] + \frac{eB}{2}[y, p_y] - \left(\frac{eB}{2}\right)^2 \underbrace{[y, x]}_{\to 0}$

$= i\hbar eB = i\hbar m\omega_c$

$\boxed{[\pi_x, \pi_y] = i\hbar m\omega_c}$ — (iv)

$\therefore [a, a^\dagger] = AA^\dagger[\pi_x + i\pi_y, \; \pi_x - i\pi_y]$

$= AA^\dagger([\pi_x, \pi_x] - i[\pi_x, \pi_y] + i[\pi_y, \pi_x])$

$= AA^\dagger(\hbar m\omega_c + \hbar m\omega_c) = 2\hbar m\omega_c AA^\dagger$

$[a, a^\dagger] = 1 \Rightarrow AA^\dagger = \frac{1}{2\hbar m\omega_c}$

$\Rightarrow |A| = \frac{1}{\sqrt{2\hbar m\omega_c}}$ $[A \in \mathbb{R}]$.

$\therefore a = \frac{1}{\sqrt{2\hbar m\omega_c}}[\pi_x + i\pi_y]$; $a^\dagger = \frac{1}{\sqrt{2\hbar m\omega_c}}[\pi_x - i\pi_y]$ (Ans) — (v)

$a^\dagger a = \frac{1}{2\hbar m\omega_c}(\pi_x^2 + i\pi_x\pi_y - i\pi_y\pi_x + \pi_y^2)$

$= \frac{1}{2\hbar m\omega_c}(\pi_x^2 + \pi_y^2 - \hbar m\omega_c)$

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$$\therefore \quad \hbar\omega_c a^\dagger a = \frac{1}{2m}(\pi_x^2 + \pi_y^2) + - \frac{\hbar\omega_c}{2}$$

$$\Rightarrow \quad \boxed{H = \hbar\omega_c \left(a^\dagger a + \frac{1}{2}\right)} \quad \text{— (VI)}$$

(III) $E_n = \left(n + \frac{1}{2}\right)\hbar\omega_c$ like in SHO.

(IV) $\pi_x = -i\hbar \frac{\partial}{\partial x} + \frac{eB}{2} y = p_x + \frac{eB}{2} y$

$\pi_y = -i\hbar \frac{\partial}{\partial y} - \frac{eB}{2} x = p_y - \frac{eB}{2} x$

$$\therefore \quad x = \frac{r + \bar{r}}{2} \; ; \quad y = \frac{\bar{r} - r}{2i} = \frac{i}{2}(r - \bar{r})$$

$$\therefore \quad \frac{\partial}{\partial x} = \partial + \bar{\partial} \; ; \quad \frac{\partial}{\partial y} = \frac{\bar{\partial} - \partial}{i} = i(\partial - \bar{\partial})$$

$$\therefore \quad \pi_x = -i\hbar(\partial + \bar{\partial}) + \frac{eB}{4} i (r - \bar{r})$$

$$\pi_y = i\hbar(\partial - \bar{\partial}) - \frac{eB}{4}(r + \bar{r})$$

$$\therefore \quad a = \frac{1}{\sqrt{2\hbar m\omega_c}} \left[-i\hbar(\partial + \bar{\partial}) + \frac{eB}{4} i(r - \bar{r}) + i\hbar(\partial - \bar{\partial}) - \frac{eBi}{4}(r + \bar{r}) \right]$$

$$= \frac{-1}{\sqrt{2\hbar m\omega_c}} \left[2i\hbar\bar{\partial} + \frac{eBi}{2}\bar{r} \right]$$

$$\boxed{a = -\frac{i}{\sqrt{2\hbar m\omega_c}} \left(2\hbar\bar{\partial} + \frac{eB}{2}\bar{r}\right)} \quad \text{— (VII)}$$

$$\therefore \quad a^\dagger = \frac{+1}{\sqrt{2\hbar m\omega_c}} \left[-i\hbar(\partial + \bar{\partial}) + \frac{eB}{4} i(r - \bar{r}) - i\hbar(\partial - \bar{\partial}) + \frac{eB}{4} i(r + \bar{r}) \right]$$

$$\boxed{a^\dagger = \frac{i}{\sqrt{2\hbar m\omega_c}} \left[-2\hbar\partial + \frac{eB}{2} r \right]} \quad \text{— (VIII)}$$

NOTE:

$$\bar{\partial}^\dagger = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)^\dagger = \frac{1}{2}\left(-\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right) = -\partial. \qquad \left[\frac{\partial}{\partial y}^\dagger = -\frac{\partial}{\partial y}\right]$$

$$\bar{r}^\dagger = r$$

$$\therefore \quad a^\dagger = \frac{i}{\sqrt{2\hbar m\omega_c}} \left[-2\hbar\partial + \frac{eB}{2} r \right] \quad [\text{from (VIII)}]$$

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Ground state condition:

$$a\psi_0 = 0.$$

$$\Rightarrow \left(2\hbar\bar{\partial} + \frac{eB}{2}\bar{r}\right)\psi_0 = 0$$

$$\Rightarrow \bar{\partial}\psi_0 = -\frac{eB}{4\hbar}\bar{r}\psi_0.$$

$$\Rightarrow \frac{\bar{\partial}\psi_0}{\psi_0} = -\frac{eB\bar{r}}{4\hbar}$$

$$\cancel{\Rightarrow \ln\psi_0 = -\frac{eB}{4\hbar}|\bar{r}|^2}$$

$$\Rightarrow \ln\psi_0 = -\frac{eB}{4\hbar}\bar{r}r + c \quad [\text{Integrate wrt } r]$$

$$\Rightarrow \boxed{\psi_0 = f(\bar{r})\exp\left(-\frac{eB}{4\hbar}r\bar{r}\right)} \quad \text{— (IX)}$$

For simplest Landau-level ground state:

$$\psi_0(r,\bar{r}) = c\exp\left(-\frac{eB}{4\hbar}r\bar{r}\right) \quad (\text{Ans})$$

NOTE: $\bar{\partial}(r\bar{r}) = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)(x^2+y^2)$

$$= \frac{1}{2}[2x + i2y] = x+iy = \bar{r}$$

$$\therefore \boxed{\bar{\partial}(r\bar{r}) = \bar{r}}$$

$$\boxed{\bar{\partial}(r^2) = 2r}$$

$$\bar{\partial}(r^2) = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)(x^2+y^2-2iy)$$

$$= \frac{1}{2}(2x + i2y - 2iy + 2x)$$

$$= (x+r+r) = 2r$$

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Thus, when acting $\bar{\partial}$ on $f(r, F)$, ~~acts~~ it only differentiates r & F acts as a constant. That is why we integrate wrt r.

(v) $$|\psi|^2 = |C|^2 \exp\left(-\frac{eB}{2\hbar} rF\right)$$

$$= |C|^2 \exp\left[-\frac{eB}{2\hbar}(x^2+y^2)\right]$$

It is symmetric in x & y. (i) It is a 2D. Gaussian centered @ the origin @ $r=0$. (ii) It decays radially outward. (iii) No angular dependence

(vi) $$\tilde{\pi}_x = p_x + eA_x.$$

$$\tilde{\pi}_x = -i\hbar\frac{\partial}{\partial x} - \frac{eB}{2}y. = p_x - \frac{eB}{2}y$$

$$\tilde{\pi}_y = -i\hbar\frac{\partial}{\partial y} + \frac{eB}{2}x = p_y + \frac{eB}{2}x$$

$$\therefore [\tilde{\pi}_x, \tilde{\pi}_y] = \left[p_x - \frac{eB}{2}y,\ p_y + \frac{eB}{2}x\right].$$

$$= \frac{eB}{2}[p_x, x] - \frac{eB}{2}[y, p_y]$$

$$= -i\hbar eB = -i\hbar m\omega_c$$

$$\therefore \boxed{[\tilde{\pi}_x, \tilde{\pi}_y] = -i\hbar eB}$$

This gives $-$ sign unlike $[\pi_x, \pi_y]$. Hence

$$b = B[\tilde{\pi}_x - i\tilde{\pi}_y] \quad \therefore \quad b^\dagger = B^\dagger[\tilde{\pi}_x + i\tilde{\pi}_y]$$

$$\therefore [b, b^\dagger] = BB^\dagger\left[i[\tilde{\pi}_x, \tilde{\pi}_y] - i[\tilde{\pi}_y, \tilde{\pi}_x]\right]$$

$$= BB^\dagger(2\hbar eB) = 1 \quad \Rightarrow \quad \boxed{|B| = \frac{1}{\sqrt{2\hbar m\omega_c}}}$$

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$\therefore\ b = \frac{1}{\sqrt{2\hbar m\omega_c}}(\tilde{\pi}_x - i\tilde{\pi}_y)$; $b^\dagger = \frac{1}{\sqrt{2\hbar m\omega_c}}(\tilde{\pi}_x + i\tilde{\pi}_y)$ (Ans)

$$[H, b] = \left(\frac{1}{2m}\cdot\frac{1}{\sqrt{2\hbar m\omega_c}}\right)[\pi_x^2 + \pi_y^2, \tilde{\pi}_x - i\tilde{\pi}_y].$$

$$= p\left([\pi_x^2, \tilde{\pi}_x] - i[\pi_x^2, \tilde{\pi}_y] + i[\pi_y^2, \tilde{\pi}_x] - i[\pi_y^2, \tilde{\pi}_y]\right).$$

$$[\pi_x, \tilde{\pi}_x] = [p_x + \frac{eB}{2}y, \cancel{p_y - e} \; p_x - \frac{eB}{2}y] = 0 = [\pi_y, \tilde{\pi}_y]$$

$$\cancel{= -\frac{eB}{2}[p_x,}$$

$$\therefore [\pi_x, \tilde{\pi}_y] = [p_x + \frac{eB}{2}y, p_y + \frac{eB}{2}x].$$

$$= \frac{eB}{2}[p_x, x] + \frac{eB}{2}[y, p_y].$$

$$= \frac{eB}{2}[-i\hbar + i\hbar] = 0.$$

$$\therefore [\pi_y, \tilde{\pi}_x] = [p_y - \frac{eB}{2}x, p_x - \frac{eB}{2}y]$$

$$= -\frac{eB}{2}[p_y, y] - \frac{eB}{2}[x, p_x]$$

$$= -\frac{eB}{2}[-i\hbar + i\hbar] = 0.$$

$$\therefore \boxed{[H, b] = 0}$$

$$\therefore [a, b] = q[\pi_x + i\pi_y, \tilde{\pi}_x - i\tilde{\pi}_y] = 0$$

[Proved]

$$[a^\dagger, b] = q[\pi_x - i\pi_y, \tilde{\pi}_x - i\tilde{\pi}_y] = 0$$

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As can be seen from calculations, $[H, b^\dagger] = 0$.

Thus, for an eigenstate $|n\rangle$ of H.

$$H|n\rangle = E_n |n\rangle.$$

$$H(b^\dagger|n\rangle) = b^\dagger(H|n\rangle) = E_n(b^\dagger|n\rangle).$$

$\therefore$ $b^\dagger|n\rangle$ is another eigenstate of H with same energy E_n. $b^\dagger|n\rangle$ & $|n\rangle$ are degenerate eigenstates.

Repeated applicat^n of $b^\dagger$ on $|n\rangle$ generates a family of degenerate eigenstates.

(iii) The conserved quantity is L_z:

$$L_z = xp_y - yp_x$$

$$H = \frac{1}{2m}[\pi_x^2 + \pi_y^2] \quad ; \quad \bar{A} = \frac{B}{2}(-y, x, 0) \quad ; \quad \bar{B} = B\hat{k}$$

Thus, H is invariant under rotation about z-axis (as is $\bar{A}$ & $\bar{B}$). The generator of rotations is conserved.

$L_z \rightarrow$ generator of rotations in 2D.

$$\therefore [H, L_z] = 0 \longrightarrow L_z \text{ is conserved}$$

L_z labels the degenerate eigenstates of a Landau-level. The ∞ degeneracy comes from conservat^n of L_z.